

Ziyi Liu

TEA Lab
HIS, Tsinghua University

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EDUCATION

University of Calgary , Alberta, Canada M.Sc. in Electrical Engineering, GPA: 3.85/4.0	Sept. 2020 - Oct. 2022
Zhejiang University of Technology , Zhejiang, China B.Eng. in Software Engineering, GPA: 3.72/5.0 (top 5%)	Sept. 2016 - Jun. 2020

RESEARCH EXPERIENCE

TEA Lab, Tsinghua University Visiting Researcher with Prof. Huazhe Xu - A drawing-based benchmark for evaluating fine-grained action generation in VLA models.	Nov. 2025 - Current
RESL Lab, University of Southern California Visiting Researcher with Prof. Gaurav Sukhatme - Few-shot imitation learning via employing multi-task datasets for knowledge transfer	Sep. 2023 - Sep. 2025
CLVR Lab, University of Southern California Visiting Researcher with Prof. Joseph J. Lim - Automated long-horizon task learning in robotics guided by large language models	Nov. 2022 - Oct. 2023
I2Sense Lab, University of Calgary Research Assistant with Prof. Orly Yadid-Pecht - Lightweight neural networks for extreme dual-exposure image fusion	Sept. 2020 - Oct. 2022
I2Sense Lab, University of Calgary Visiting Undergraduate Researcher with Prof. Orly Yadid-Pecht - Image tone mapping through a reformulated laplacian neural network	Apr. 2019 - Apr. 2020

IN SUBMISSION

[S1] Xiangyu Xu*, **Ziyi Liu***, Bangshuai Peng*, Mengjie Zhao*, Chenhao Lu, Zhengyang Fan, Kaizhe Hu, Huazhe Xu. “BURGER: Towards Benchmarking Unlimited Robotic Generalization via Embodied dRawing”, *CoRL 2026 Submission*, 2026.

CONFERENCE PAPERS

[C1] Jesse Zhang, Jiahui Zhang, Karl Pertsch, **Ziyi Liu**, Xiang Ren, Minsuk Chang, Shao-Hua Sun, and Joseph J Lim. “Bootstrap Your Own Skills: Learning to Solve New Tasks with Large Language Model Guidance”, *Oral at CoRL 2023 (top 6.6%). Spotlight at Articulate Robots Workshop at RSS 2023*, 2023

[C2] **Ziyi Liu***, Grace Zhang*. “Generalize and Guide: Decomposing Rewards for Few-Shot Inverse Reinforcement Learning”, *RLC*, 2026. *RSS, RC4Robotics Workshop*, 2025.

PREPRINTS

- [P5] **Ziyi Liu**, Jie Yang, Svetlana Yanushkevich, and Orly Yadid-Pecht. “LightFuse: Lightweight CNN based Dual-exposure Fusion”, *ArXiv Preprint*, 2021
- [P4] **Ziyi Liu**, Jie Yang, Mengchen Lin, Kenneth Kam Fai Lai, Svetlana Yanushkevich, and Orly Yadid-Pecht. “WDR FACE: The First Database for Studying Face Detection in Wide Dynamic Range”, *ArXiv Preprint*, 2020
- [P3] Jie Yang, **Ziyi Liu**, Mengchen Lin, Svetlana Yanushkevich, and Orly Yadid-Pecht. “Deep reformulated Laplacian tone mapping”, *ArXiv Preprint*, 2019
- [P2] Jie Yang, **Ziyi Liu**, Ulian Shahnovich, and Orly Yadid-Pecht. “Tone Mapping Based on Multi-scale Histogram Synthesis”, *ArXiv Preprint*, 2019
- [P1] Jie Yang, Mengchen Lin, **Ziyi Liu**, Ulian Shahnovich, and Orly Yadid-Pecht. “Mobile-end Tone Mapping based on Integral Image and Integral Histogram”, *ArXiv Preprint*, 2019

HONORS AND AWARDS

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| • International Graduate Tuition Award | Oct. 2021 |
| • Graduate Fellowship | Sep. 2020 |
| • Mitacs Globalink Research Scholarship | Jan. 2019 |
| • Third Prize of Coding Competition of Zhejiang University of Technology | Mar. 2018 |
| • Zhejiang Provincial Government Scholarship (top 3%) | Nov. 2017 |

LEADERSHIP AND ACTIVITIES

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| • Reviewer, CoRL 2026 | July. 2026 |
| • Student Volunteer, CoRL 2023 | Nov. 2023 |
| • Operator, From Lab 2 Fulfillment | Sep. 2020 - Oct. 2022 |
| • Volunteer, Women in Science and Engineering | Sep. 2019 |
| • Team Leader, ACM China Collegiate Programming Contest | Oct. 2018 |

COMPUTER SKILLS

Programming Languages: Python, TensorFlow, Keras, PyTorch, C/C++, Java, SQL, MATLAB, C#

SDKs & Tools: PyCharm, Eclipse, Visual Studio, Git, Linux Shell, .Net, Vivado, **AI2-THOR**